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United States Patent	12414874
Kind Code	B2
Date of Patent	September 16, 2025
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Eye drop dispenser

Abstract

A dispenser assembly includes a dispenser bottle having a nipple for dispensing a solution (e.g., eye drop solution, eye wash solution, eye lid cleaner and/or wash solution), and a cap removably couplable to the dispenser bottle over the nipple. An eye cup is fixedly coupled to the dispenser bottle around the nipple. The eye cup can be supported on or rest on the periphery of the eye socket to facilitate delivery of the solution onto or over a user's eye.

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Family ID:	1000008822172
Appl. No.:	18/800555
Filed:	August 12, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20250186254 A1	Jun. 12, 2025

Related U.S. Application Data

continuation parent-doc US 18535817 20231211 US 12070415 child-doc US 18800555

Publication Classification

Int. Cl.: A61F9/00 (20060101)

U.S. Cl.:

CPC A61F9/0026 (20130101);

Field of Classification Search

CPC: A61F (9/0008); A61F (9/0026); A61F (2250/0097); A61F (2230/0063); B65D (47/18); B65D (1/0223); B65D (1/08); B65D (47/2031); A61H (35/02); A61H (2201/0153); A61H (2033/048); A61M (2210/0612); A61M (11/008); A61L (12/086)

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Background/Summary

INCORPORATION BY REFERENCE TO ANY PRIORITY APPLICATIONS (1) Any and all applications for which a foreign or domestic priority claim is identified in the Application Data

Sheet as filed with the present application are hereby incorporated by reference under 37 CFR 1.57. This application is a continuation of U.S. patent application Ser. No. 18/535,817, filed on Dec. 11, 2023, the entire disclosure of which is incorporated herein by reference.

FIELD

(1) The present disclosure is directed to an eye drop or eye wash or eye lid cleaner dispenser bottle, and more particularly to an eye drop or eye wash or eye lid cleaner dispenser with an eye cup for dispensing eye drops or eye wash or eye lid cleaner.

DESCRIPTION OF THE RELATED ART

(2) Eye drops are regularly used to treat eye conditions such as dry eye, bacterial infections, eye injury, or eye conditions such as glaucoma and cataracts. Existing eye drop dispensers are deficient in that they do not accurately or consistently deliver the drops into the eye (e.g., affecting children and elderly people more), often delivering the drops on other parts of the user's face, leading to multiple attempts and making the solution difficult to administer. These multiple attempts cause an unnecessary wasting of the eye drops, thereby costing the consumer more. Another deficiency of existing eye drop dispensers is that user's may contaminate existing eye drop dispensers by touching their eye or face with the dispenser causing bacteria to grow on the dispenser tip and leading to eye infections. When the user suffers from hand tremors (e.g., essential tremor disorder or other neurological conditions) which cause shaking hands while trying to deliver the eye drops, the tremors exacerbate the defects of existing eye drop dispensers. Because existing dispenser designs make delivering eyedrops difficult, there is a decrease in compliance by users to take their medication.

SUMMARY

(3) The systems, methods, and devices described herein have innovative aspects, no single one of which is indispensable or solely responsible for their desirable attributes. Without limiting the scope of the claims, some of the advantageous features will now be summarized.

(4) In some aspects, the techniques described herein relate to an eye drop dispenser assembly, including: a dispenser bottle having a nipple via which a solution is dispensed, the nipple projecting from a surface of the dispenser bottle, and a cap removably couplable to the dispenser bottle over the nipple. The cap having an outer cap, an inner cap extending within the outer cap, and a seal recessed in a hollow end of the inner cap. The assembly also includes an eye cup fixedly coupled to a second surface of the dispenser bottle and configured to fit within the cap when the cap is coupled to the dispenser bottle. The eye cup having a circumferential wall disposed around the nipple and a distal edge of the circumferential wall defining an oval shape. The distal edge configured to be disposed over an eye socket to facilitate delivery of the solution into a user's eye. The nipple disposed between the distal edge of the circumferential wall and the dispenser body. Wherein, when the cap is coupled to the dispenser body, the seal sits adjacent the nipple to inhibit leakage of the solution from the nipple and the distal portion of the inner cap encircles the nipple.

(5) In some aspects, the second surface of the dispenser is coplanar with the surface of the dispenser.

(6) In some configurations, a height of the nipple above the surface of the dispenser is less than half of a height of the eye cup above the second surface.

(7) In some configurations, the nipple is recessed proximal to the dispenser bottle.

(8) In some configurations, the seal comprises rubber.

(9) In some configurations, the dispenser bottle comprises plastic.

(10) In some aspects, the techniques described herein relate to an eye drop dispenser assembly, including: a dispenser body having a nipple through which a solution is dispensed, the nipple projecting from the dispenser body and a cap removably couplable to the dispenser body over the nipple. The cap having a cavity, a post extending within the cavity, and a seal recessed in a hollow end of the post. The assembly also including an eye cup fixedly coupled to the dispenser body and

disposed around the nipple. The eye cup having a circumferential wall and a distal edge configured to be disposed over an eye socket to facilitate delivery of a solution into a user's eye. The nipple disposed between the distal edge of the circumferential wall and the dispenser body. Wherein, when the cap is coupled to the dispenser body, the seal inhibits leakage of the solution from the nipple and a distal portion of the post encircles the nipple.

(11) In some configurations, the eye cup is configured to inhibit the nipple from touching the user's eye.

(12) In some configurations, the body has a recess from which the nipple projects opposite the distal edge of the eye cup.

(13) In some configurations, a height of the nipple above the dispenser body is less than half a height of the distal edge of the eye cup above the dispenser body.

(14) In some configurations, the dispenser body has a recess extending away from the distal edge of the eye cup, the nipple projecting towards the distal edge of the eye cup from the recess.

(15) In some configurations, the circumferential wall of the eye cup is partially disposed within the recess.

(16) In some configurations, a distal portion of the post extends past the seal.

(17) In some configurations, a distal portion of the post is couplable to the nipple.

(18) In some configurations, the dispenser body has a shoulder, and the cap is couplable to the dispenser body around the shoulder.

(19) In some configurations, the cap is couplable to the dispenser body with an annular snap fit.

(20) In some aspects, the techniques described herein relate to an eye drop dispenser assembly, including: a dispenser bottle having a nipple via which a solution is dispensed, the nipple projecting from a surface of the dispenser bottle and a cap removably couplable to the dispenser bottle over the nipple. The cap having an outer cap, an inner cap extending within the outer cap, and a seal recessed in the inner cap. The assembly also including an eye cup fixedly coupled to a second surface of the dispenser bottle. The eye cup having a distal edge defining an oval shape, disposed circumferential to the nipple. Wherein, when the cap is coupled to the dispenser body, the seal sits adjacent the nipple.

(21) In some configurations, the eye cup is coupled to the surface.

(22) In some configurations, the dispenser assembly is configured to prevent contamination of the nipple and inadvertent contact of the nipple with the user's eye.

(23) In some configurations, the distal edge of the eye cup is contoured to fit around a periorbital region.

(24) In some configurations, the surface of the bottle is in a recess below a shoulder of the bottle.

(25) In some configurations, the nipple is disposed within the recess.

(26) In some configurations, the distal edge is contoured relative to an axis defined by a periorbital region.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1A is a schematic front view of one embodiment of an eye drop dispenser assembly.

(2) FIG. 1B is a schematic front view of a second embodiment of an eye drop dispenser assembly.

(3) FIG. 2A is a schematic front view of the eye drop dispenser assembly of FIG. 1A with its cap removed and spaced above the dispenser bottle, the dashed line showing the outline of the cap when covering the eye cup.

(4) FIG. 2B is a schematic side view of the eye drop dispenser assembly of FIG. 1A with its cap removed and spaced above the dispenser bottle.

(5) FIG. 2C is a schematic cross-sectional view of the eye drop dispenser assembly of FIG. 1A,

where the section line is through the midplane of the side view in FIG. 2B.

(6) FIG. 3A is a schematic front view of the eye drop dispenser assembly of FIG. 1B with its cap removed and spaced above the dispenser bottle, the dashed line showing the outline of the cap when covering the eye cup.

(7) FIG. 3B is a schematic side view of the eye drop dispenser assembly of FIG. 1B with its cap removed and spaced above the dispenser bottle.

(8) FIG. 3C is a schematic cross-sectional view of the eye drop dispenser assembly of FIG. 1B, where the section line is through the midplane of the side view in FIG. 3B.

(9) FIG. 4A is a schematic top view of the eye drop dispenser assembly of FIG. 1A.

(10) FIG. 4B is a schematic top view of the eye drop dispenser assembly of FIG. 1B.

(11) FIG. 5A is a schematic top view of the cap of the eye drop dispenser assembly of FIG. 1A and FIG. 1B.

(12) FIG. 5B is a schematic bottom view of the cap of FIG. 5A.

(13) FIG. 5C is a schematic cross-sectional view of the front view of one embodiment of the cap of FIG. 5A, where the section line is through the midplane of the side view.

(14) FIG. 5D is a schematic cross-sectional view of the front view of a second embodiment of the cap of FIG. 5A, where the section line is through the midplane of the side view.

(15) FIG. 6 is a schematic view of the eye drop dispenser of FIG. 1A in an inverted position over a user's eye to deliver an eye drop thereto.

DETAILED DESCRIPTION

(16) Embodiments of the eye drop dispenser assembly will now be described with reference to the accompanying figures, wherein like numerals refer to like or similar elements throughout. Although several embodiments, examples, and illustrations are disclosed below, it will be understood by those of ordinary skill in the art that the inventions described herein extends beyond the specifically disclosed embodiments, examples, and illustrations, and can include other uses of the inventions and obvious modifications and equivalents thereof. The terminology used in the description presented herein is not intended to be interpreted in any limited or restrictive manner simply because it is being used in conjunction with a detailed description of certain specific embodiments of the inventions. In addition, embodiments of the inventions can comprise several novel features and no single feature is solely responsible for its desirable attributes or is essential to practicing the inventions herein described.

(17) FIGS. 1A, 2A-2C, 4A, and 5C show a dispenser assembly **100** (the “assembly **100**”). The assembly **100** includes a dispenser bottle **150** (e.g., dispenser body), an eye cup **140**, a nipple **130**, and a cap **110**. The dispenser bottle **150**, the eye cup **140**, and/or the cap **110** can optionally be made in part or whole of plastic (e.g., polypropylene), silicone or rubber. However, the dispenser bottle **150**, the eye cup **140**, and/or the cap **110** can be made of other suitable materials (e.g., flexible or resilient materials). The eye cup **140** may be formed from a clear or transparent material. The dispenser assembly **100** can house and deliver a solution (e.g., eye drops, eye wash, eye lid cleaner and/or wash) onto or over the eye or a portion thereof (e.g., eye lid).

(18) FIGS. 1B, 3A-3C, 4B, and 5D show views of a dispenser assembly **100'** (hereafter “the assembly **100'**”), which is a compact (e.g., small or pocket sized) version of the dispenser assembly **100**. Some of the features of the assembly **100'** are similar to features of the assembly **100** in FIGS. 1A, 2A-2C, 4A, and 5C. Thus, reference numerals used to designate the various components of the assembly **100'** are identical to those used for identifying the corresponding components of the assembly **100** in FIGS. 1A, 2A-2C, 4A, and 5C, except that a “'” has been added to the numerical identifier. Therefore, the structure and description for the various features of the assembly **100** and how it is operated and controlled in FIGS. 1A, 2A-2C, 4A, and 5C are understood to also apply to the corresponding features of the assembly **100'**, except as described below. The dispenser assembly **100'** can house and deliver a solution (e.g., eye drops, eye wash, eye lid cleaner and/or wash) onto or over the eye or a portion thereof (e.g., eye lid).

(19) The dispenser assembly **100** includes a dispenser bottle **150** with a base **152** and a cap **110** removably couplable to an end of the dispenser bottle **150** opposite to (e.g., distal from) the base **152**. In some embodiments, the base **152** is flat. In other embodiments, a portion, but less than all, of the base **152** is flat. The base **152** allows the dispenser bottle **150** to stand upright when not in use which advantageously reduces the amount of fluid (e.g., eye drops, eye wash, eye lid cleaner and/or wash) wasted. The cap **110** is removably coupled to the dispenser bottle **150**. The assembly **100** has a height A measured from the base **152** to a top of the cap **110**. In one implementation, the height A is between about 2 and 3 inches, such as 2.5 inches.

(20) The dispenser assembly **100'** includes a dispenser bottle **150'** (e.g., dispenser body) with a base **152'** and a cap **110'** removably couplable to an end of the dispenser bottle **150'** opposite to (e.g., distal from) the base **152'**. In some embodiments, the base **152'** is flat. In other embodiments, a portion, but less than all, of the base **152'** is flat. The assembly **100'** has a height A' measured from the base **152'** to a top of the cap **110'**. In one implementation, the height A' is between about 1.5 and 2.5 inches, such as 2 inches.

(21) FIG. 2A depicts a front view of the dispenser assembly **100** shown in FIG. 1A with the cap **110** removed. The dispenser bottle **150** has the base **152** connected to a side wall **158**. The side wall **158** is circumferential and extends from the base **152**. The dispenser bottle **150** has a nipple **130** projecting from a surface **164** of the dispenser bottle **150** that is at an opposite end of the dispenser bottle **150** (e.g., distal) from the base **152**. The nipple **130** can have a circular profile, as shown for example in FIG. 4A; however, the nipple **130** can have other suitable shapes. In some embodiments, the circumferential side wall **158** can flex (e.g., when the dispenser bottle **150** is squeezed). The dispenser bottle **150** also has a shoulder **154** that circumscribes a second surface **160**. The second surface **160** connect to the circumferential side wall **158**. In the illustrated embodiment the surface **164** is coplanar with the second surface **160**. The cap **110** is removably couplable to the dispenser bottle **150** around the shoulder **154**. In some examples the cap **110** is removably couplable to the shoulder **154** with an annular snap fit. In some embodiments, the shoulder **154** is disposed to extend above the second surface **160**. In other embodiments, the shoulder **154** is formed integral to the second surface **160** (e.g., the second surface **160** forms the upper surface of the shoulder **154** such that when the cap **110** is on, or connects to, the second surface **160** fits within the cap **110** rather than the cap **110** sitting on the second surface **160**). In the illustrated arrangement, the assembly **100** also includes an eye cup **140**, which provides a supporting structure to the dispenser bottle **150** when it is inverted for delivery of solution on or over the eye (e.g., the eye cup **140** resting on the user's eye socket to support the dispenser bottle **150** in the inverted position). The eye cup **140** is coupled to the second surface **160** of the dispenser bottle **150** (e.g., to an end of the dispenser bottle **150** opposite the base **152**). The eye cup **140** can fit within the cap **110** when the cap **110** is coupled to the dispenser bottle **150**. In some embodiments, the eye cup **140** is fixedly coupled to the second surface **160** (e.g., so that the eye cup **140** is integral to, and cannot be detached or removed from, the dispenser bottle **150**). The eye cup **140** has a circumferential wall **142** that extends to an edge **144** that is distal to the dispenser bottle **150**. In one example, the eye cup **140** (e.g., the circumferential wall **142**) tapers toward the second surface **160**. The circumferential wall **142** defines an oval shape surrounding the nipple **130** (e.g., when the dispenser assembly **100** is viewed from the top with the cap **110** removed, see FIG. 4A. The circumferential wall **142** surrounds the nipple **130**, and the nipple **130** is disposed between the distal edge **144** of the circumferential wall and the second surface **160** of the dispenser bottle **150**.

(22) The dispenser bottle **150** can optionally be made of plastic, silicone, rubber, or other suitable resilient or flexible material. The dispenser bottle **150** can optionally be a single piece (e.g., a monolithic, seamless single piece). In some embodiments, the base **152**, side wall **158**, surface **164**, second surface **160**, shoulder **154**, nipple **130**, and eye cup **140**, or any combination thereof, may be formed as a single piece. The single piece can be an integrated assembly of several parts or can be a

unitary structure formed from a single piece of material. In some embodiments, the base **152**, side wall **158**, surface **164**, second surface **160**, shoulder **154**, nipple **130**, and eye cup **140** may be made from plastic, rubber, silicone or any suitable material. Any of these portions may be made from the same material or from different materials.

(23) In the illustrated arrangement, the assembly **100** has a width C. In one implementation, the width C is between 1.5 and 2.5 inches, such as 2 inches. The dispenser bottle **150** has a height B measured from the base **152** to the surface **164**. In one implementation, the height B is between 1.25 and 2 inches, such as 1.625 inches. The dispenser bottle **150** also has a height I measured from the surface **164** to the top of the nipple **130**. The assembly **100** has a height F measured from the second surface **160** to the distal edge **144**. Height F is advantageously larger than height I to inhibit (e.g., prevent) the nipple **130** from contacting the eye **200** when in use. This positioning advantageously prevents (e.g., inhibits) contamination of the nipple **130** or injury to the eye **200** during use. Additionally, in some examples the height I may be less than half of height F which advantageously further reduces the chance that the nipple **130** will inadvertently touch the user's eye **200** or otherwise be contaminated. In one implementation, the height I is between 0 and 0.4 inches, such as 0.2 inches and the height F is between 0.25 and 0.75 inches, such as 0.5 inches. The assembly **100** also has a height D. Height D is the height of the cap **110**. In one implementation, height D is greater than height F. In one implementation, the height D is between 0.5 and 1.0 inches, such as 0.875 inches.

(24) With reference to FIG. 2B, the assembly **100** has a depth G. The base **152** has a depth H. Depth G may be larger than depth H. In one implementation, the depth G is between 1 and 1.5 inches, such as 1.25 inches and the depth H is between 0.5 and 1 inch, such as 0.75 inches. The eye cup **140** has a depth J. Depth J may be smaller than Depth G. In one implementation, the depth J is between 1 and 1.25 inches, such as 1.1 inches. The cap **110** has a depth K (at its distal end). In one implementation, the depth K is between 0.5 and 0.75 inches, such as 0.7 inches.

(25) FIG. 2C depicts a cross-sectional view of the dispenser assembly **100** shown in FIG. 2A. The section is taken at a midpoint of the depth G of the assembly **100** (see FIG. 2B). The dispenser bottle **150** has a chamber **162** that can hold a liquid solution (e.g., eye drop solution, eye wash solution, eye lid cleaner or wash solution). The cap **110** is connected to the dispenser bottle **150**. The eye cup **140** fits within a cavity **112** of the cap **110** when the cap **110** is removably coupled to the dispenser bottle **150**. The cap **110** has an outer cap **114** that defines the cavity **112**. Within the cavity **112**, is a post **118** extending from an underside of the outer cap **114**. In one example, the post **118** is solid. In another example, the post **118** includes a hollow portion (e.g., at a distal end of the post **118**). The post **118** has a seal **120** at its distal end. In one example, the seal **120** is separate from the post **118** and is disposed within the hollow portion of the post **118**. In another example, the seal **120** is separate from the post **118** and is attached to an end of the post **118**. In another example, the seal **120** is integral (e.g., one piece, monolithic) with the post **118** (e.g., the seal **120** is a distal portion of the post **118** and made of the same material as the post **118**). When the cap **110** is coupled to the dispenser bottle **150**, the outer cap **114** is coupled to (e.g., disposed around) the shoulder **154** of the dispenser bottle **150**. The post **118** and the seal **120** define an inner cap **116** that is coupled to (e.g., disposed around) the nipple **130** so that a distal portion of the post **118** encircles the nipple **130** and is coupled to the nipple **130**. The seal **120** sits adjacent the top of the nipple **130** and presses against the nipple **130** to inhibit (e.g., prevent) leakage of solution from the nipple **130**. In another example, when the cap **110** is coupled to the dispenser bottle **150**, the seal **120** at the end of the post **118** engages (e.g., contacts) the nipple **130** to inhibit (e.g., prevent) leakage of solution from the dispenser bottle **150**. In one example, the seal **120** can have a shape similar to (e.g., a mirror image of) a shape of the nipple **130**). The post **118** is disposed within the eye cup **140** when the cap **110** is connected to the dispenser bottle **150** (e.g., when the cap **110** is coupled to the dispenser bottle **150** the post **118** is surrounded by the eye cup **140**, which is surrounded by the outer cap **114**).

(26) FIG. 3A depicts a front view of the dispenser assembly **100'** shown in FIG. 1B with the cap **110'** removed. The assembly **100'** has a dispenser bottle **150** comprising a base **152'** connected to a side wall **158'**. The dispenser bottle **150'** has a nipple **130'** projecting from a surface **164'** (e.g., a recessed surface) of the dispenser bottle **150'** that is at an opposite end of the dispenser bottle **150'** (e.g., distal) from a base **152'**. The nipple **130'** can have a circular profile, as shown for example in FIG. 4B; however, the nipple **130'** can have other suitable shapes. The dispenser bottle **150'** also has a shoulder **154'** circumferential to a second surface **160'** and distal to the base **152'**, as shown in FIG. 3A. The side wall **158'** also connected to the second surface **160'**. In some embodiments, the shoulder **154'** is disposed to extend above the second surface **160'**. In other embodiments, the shoulder **154'** is formed integral to the second surface **160'** (e.g., the second surface **160'** forms the upper surface of the shoulder **154'** such that when the cap **110'** is on, the second surface **160'** fits within the cap **110'** rather than the cap **110'** sitting on the second surface **160'**). The second surface **160'** is parallel to the base **152'**. In the illustrated arrangement, shown in FIG. 3C, the assembly **100'** also includes an eye cup **140'**. The eye cup **140'** is fixedly coupled to the dispenser bottle **150'**. When the cap **110'** is coupled to the dispenser bottle **150'**, the eye cup **140'** can fit within the cap **110'**. The eye cup **140'** has a circumferential wall **142'** and an edge **144'** (e.g., a top edge) that is distal to the dispenser bottle **150'**. In one example, the eye cup **140'** (e.g., the circumferential wall **142'**) tapers toward the second surface **160'**.

(27) With reference to FIG. 3A, the assembly **100'** has a width C'. In one implementation, the width C' is between 1.5 and 2.5 inches, such as 1.9 inches.

(28) With reference to FIG. 3B, the assembly **100'** has a depth G'. The base **152'** has a depth H'. Depth G' may be larger than depth H'. In one implementation, the depth G' is between 1 and 1.5 inches, such as 1.2 inches and the depth H' is between 0.5 and 1 inches, such as 0.8 inches. The eye cup **140'** has a depth J'. Depth J' may be smaller than Depth G'. In one implementation, the depth J' is between 1 and 1.25 inches, such as 1.1 inches. The cap **110'** has a depth K' (at its distal end). In one implementation, the depth K' is between 0.5 and 1 inches, such as 0.9 inches.

(29) FIG. 3C depicts a cross-sectional view of the dispenser assembly **100'** shown in FIG. 3A. The section is taken at a midpoint of the depth G' of the assembly **100'** (see FIG. 3B). The dispenser bottle **150'** has a chamber **162'** that can hold a liquid. The cap **110'** is connected to the dispenser bottle **150'**. The dispenser bottle **150'** has a nipple **130'** projecting from a surface **164'**. The surface **164'** is between the second surface **160'** and the base **152'**. The positioning of the surface **164'** creates a recess **156'** around the nipple **130'**. The recess **156'** extends into the dispenser bottle **150'** and away from the distal edge **144'** of the eye cup **140'**. The positioning of the recess **156'** allows the nipple **130'** to be recessed proximal (e.g., the nipple **130'** is nested in the recess **156'** of the dispenser bottle **150'**) to the dispenser bottle **150'** while projecting towards the distal edge **144'** of the eye cup **140'**, allowing the dispenser assembly **100'** to be more compact than the dispenser assembly **100**. The eye cup **140'** fits within a cavity **112'** of the cap **110'** and the circumferential wall **142'** defines an oval shape surrounding the nipple **130'**. The nipple **130'** is disposed between distal edge **144'** of the circumferential wall **142'** and the surface **164'** of the dispenser bottle **150'**. In one example, the circumferential wall **142'** is partially disposed within the recess **156'**. The cap **110'** has an outer cap **114'** that defines the cavity **112'**. Within the cavity **112'**, is a post **118'** extending from the outer cap **114'**. In one example, the post **118'** is solid. In another example, the post **118'** includes a hollow portion (e.g., at a distal end of the post **118'**). The post **118'** has a seal **120'** at its distal end. In one example, the seal **120'** is separate from the post **118'** and is disposed within the hollow portion (e.g., distal hollow portion) of the post **118'**. In another example, the seal **120'** is separate from the post **118'** and is attached to an end of the post **118'**. In another example, the seal **120'** is integral (e.g., one piece, monolithic) with the post **118'** (e.g., the seal **120'** is a distal portion of the post **118'** and made of the same material as the post **118'**). The outer cap **114'** is disposed around the shoulder **154'** of the bottle **150'**. The post **118'** and the seal **120'** define an inner cap **116'**. The seal **120'** sits adjacent the top of the nipple **130'** and presses against the nipple **130'** to inhibit

leakage of a solution or liquid from the nipple **130'**. In one example, the seal **120'** can have a shape similar to (e.g., a mirror image of) a shape of the nipple **130'**). The post **118'** is disposed within the eye cup **140'** when the cap **110'** is connected to the dispenser bottle **150'**.

(30) In the illustrated arrangement, the bottle **150'** has a height **B'** measured from the base **152'** to the second surface **160'**. In one implementation, the height **B'** is between 1 and 1.5 inches, such as 1.25 inches. The assembly **100'** has a height **F'** (for the eye cup **140'**) measured from the surface **164'** to the distal edge **144'**. In one example, the eye cup **140'** extends from the second surface **160'** (e.g., the eye cup **140'** is outside the recess **156'**). In another example, the eye cup **140'** extends from the surface **164'** so that at least a portion of the circumferential wall **142'** of the eye cup **140'** is in the recess **156'** of the bottle **150'**. In one implementation, the height **F'** is between 0.25 and 0.75 inches, such as 0.4 inches. The assembly **100'** also has a height **D'**. Height **D'** is the height of the cap **110'**. The height **D'** is greater than height **F'**. In one implementation, the height **D'** is between 0.5 and 1 inches, such as 0.75 inches. The dispenser bottle **150'** has a height **E'** measured from the second surface **160'** to the surface **164'**. In one implementation, the height **E'** is between 0.1 and 0.4 inches, such as 0.25 inches. In some arrangements, the second surface **160'** may be coplanar with the surface **164'** and height **E'** is zero inches. In some examples, the second surface **160'** and the surface **164'** can be portions of the same surface.

(31) FIG. 4A depicts a top view of the assembly **100**. The dispenser bottle **150** is shown with the distal edge **144** of the eye cup **140** comprising an oval shape and disposed around the nipple **130**. In this embodiment, the dispenser bottle **150**, has an oval shaped perimeter. In other implementations, the perimeter of the dispenser bottle **150** or eye cup **140** may take different forms (e.g. substantially polygonal, or substantially circular).

(32) FIG. 4B depicts a top view of the assembly **100'**. The dispenser bottle **150'** is shown with the distal edge **144'** of the eye cup **140'** comprising an oval shape and disposed around the nipple **130'**. In this embodiment, the dispenser bottle **150'**, has an oval shaped perimeter. In other implementations, the perimeter of the dispenser bottle **150'** or eye cup **140'** may take different forms (e.g. substantially polygonal, or substantially circular).

(33) FIGS. 5A-5D depict the cap **110**. FIG. 5A depicts a top view of the cap **110** and FIG. 5B depicts a bottom view of the cap **110**. It is understood that this illustration can be either of the embodiments shown in FIGS. 1A and 1B. FIG. 5C depicts a cross-sectional front view of one embodiment of the cap **110** and FIG. 5D depicts a cross-sectional front view of another embodiment of the cap **110**. It is understood that FIGS. 5A and 5B may depict either the embodiment in 5C or the embodiment in 5D.

(34) In the arrangement illustrated in FIG. 5A, the cap **110** has an oval shaped perimeter. In other implementations, the perimeter of the cap **110** may take different forms (e.g., substantially polygonal, or substantially circular). The perimeter of the cap **110** has width **C** and a depth **G**. In one implementation, the width **C** is between 1.5 and 2.5 inches, such as 2 inches and the depth **G** is between 1 and 1.5 inches, such as 1.25 inches. The top of the cap **110** has a substantially flat surface with an oval shaped perimeter. The perimeter has a width **J** and a depth **K**. In one implementation, the width **J** is between 1.2 and 1.6 inches, such as 1.4 inches and the depth **K** is between 0.5 and 1 inches, such as 0.7 inches.

(35) FIG. 5B depicts the bottom view of the arrangement illustrated in FIG. 5A. The cap **110** has an outer cap **114**, an inner cap **116** extending within the outer cap **114**, and a seal **120** recessed in the inner cap **116**. The outer cap **114** defining a cavity **112** and the inner cap **116** defined by a post **118**.

(36) FIG. 5C depicts a cross-sectional front view of one embodiment of the cap **110**. In this embodiment, the post **118** extends into the cavity **112** and the seal **120** sits within the post **118**. In one example, a distal portion **122** of the post **118** extends past the seal **120**.

(37) FIG. 5D depicts a cross-sectional front view of a second embodiment of the cap **110'**. In this embodiment, the post **118'** extends into the cavity **112'** and the seal **120'** sits within the post **118'**. The seal **120'** extends to the extent of the post **118'** (e.g., the seal **120'** is at the distal end **122'** of the

post **118'**) and the post **118'** extends to the extent of the cavity **112'**.

(38) In some embodiments, the outer cap **114**, the inner cap **116**, and the seal **120**, or any combination thereof, may be formed as a single piece. The single piece can be an integrated assembly of several parts or can be a unitary structure formed from a single piece of material. In some embodiments, the seal **120**, the outer cap **114**, and the inner cap **116** may be made from metal, plastic, rubber, or any suitable material. Any of these portions may be made from the same material or from different materials.

(39) As shown in FIG. 6, the assembly **100** can be inverted and disposed over an eye **200** so that the eye cup **140** is disposed over (e.g., in contact with) an eye socket **210** of the eye **200** (e.g., to that the eye cup **140** provides a supporting structure that supports the dispenser bottle **150** in an inverted position on the eye socket of the user's eye). The user can then squeeze the dispenser bottle **150** to dispense one or more drops into the eye **200**. This arrangement advantageously facilitates self-administration of a solution to a user's eye **200** and is user friendly as it does not require the user to assemble the eye cup onto the bottle. Rather, the user simply removes the cap **110** from the dispenser bottle **150**, inverts the dispenser bottle **150**, places the eye cup **140** against the eye socket of the eye, and squeezes the dispenser bottle **150** to dispense the solution. Further, the nipple **130** is advantageously centered relative to the eye cup **140** so that when the eye cup **140** is supported on the eye socket **210**, the nipple **130** is generally aligned (e.g., centered) relative to the eye **200**, further ensuring the dispensed solution will be delivered onto or over the eye **200**. Advantageously, the eye cup **140** is sized to dispose the distal edge **144** over (e.g., in contact with) the periorbital region **210** and align the nipple **130** with the center of the eye **200**. The distal edge **144** of the eye cup **140** is contoured to fit around the periorbital region **210**, advantageously allowing the eye cup **140** to comfortably sit or rest over the eye socket **210**. This positioning facilitates alignment of the nipple **130** with the eye **200** allowing for dispensing the eye drop(s) directly into the eye **200**. This positioning limits (e.g., prevents) waste from having the eye drops fall on the user's face, cheek, eyebrow, ear, etc. Though FIG. 6 depicts the use of the assembly **100**, the assembly **100'** can be used in the same manner and the advantages listed above also apply to the assembly **100'**.

(40) The contour of the distal edge **144** is defined by two axes. The first axis is defined by the nipple **130** such that the distal edge **144** is contoured circumferentially around the nipple **130**. The second axis is defined by a point in space approximating the centroid of the external curvature of a user's periorbital region **210**. The distal edge **144** is contoured relative to this axis such that the eye cup **140** fits against the user's face. The ability of the eye cup **140** to sit or rest over the periorbital region **210** advantageously facilitates use of the eye drop dispenser assembly **100** by a user with hand tremors, nervous disorders or children to dispense eye drops into the eye. Additionally, supporting (e.g., sitting or resting) the eye cup **140** over the eye socket **210** can inhibit (e.g., prevent) injury to the eye during delivery of eye drops and inhibit the nipple **130** from touching the user's eye **200**, thereby inhibiting (e.g., preventing) contamination of the eye **200**.

(41) The assembly **100'** differs from the assembly **100** in that the dispenser assembly **100'** has a smaller external envelope. This is accomplished by recessing the nipple **130'** as depicted in FIG. 3C. This allows height D' to be less than height D and height A' to be less than height A. Bottle **150** can hold a larger volume of fluid than bottle **150'**.

(42) The assembly **100**, **100'** is advantageously easy to use (e.g., it is easy to remove the cap **110**, **110'**, position the eye cup **140**, **140'** over the eye **200**, and dispense drops in the optimal location without contaminating the nipple **130**, **130'**). For example, because the eye cup **140**, **140'** is integrated (e.g., fixed) with the dispenser bottle **150**, **150'**, the user can simply remove the cap **110**, **110'**, invert the dispenser bottle **150**, **150'** and position it over (e.g., in contact with) their eye socket **210** and squeeze the dispenser bottle **150**, **150'** to dispense the solution from the dispenser bottle **150**, **150'**. Additionally, the fixed eye cup **140**, **140'** advantageously reduces the likelihood that the eye cup **140**, **140'** will be lost, use improperly, or that the nipple **130**, **130'** will be contaminated.

(43) While certain embodiments of the inventions have been described, these embodiments have

been presented by way of example only and are not intended to limit the scope of the disclosure. Indeed, the novel methods and systems described herein may be embodied in a variety of other forms. Furthermore, various omissions, substitutions and changes in the systems and methods described herein may be made without departing from the spirit of the disclosure. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosure. Accordingly, the scope of the present inventions is defined only by reference to the appended claims.

(44) Features, materials, characteristics, or groups described in conjunction with a particular aspect, embodiment, or example are to be understood to be applicable to any other aspect, embodiment or example described in this section or elsewhere in this specification unless incompatible therewith. All of the features disclosed in this specification (including any accompanying claims, abstract and drawings), and/or all of the steps of any method or process so disclosed, may be combined in any combination, except combinations where at least some of such features and/or steps are mutually exclusive. The protection is not restricted to the details of any foregoing embodiments. The protection extends to any novel one, or any novel combination, of the features disclosed in this specification (including any accompanying claims, abstract and drawings), or to any novel one, or any novel combination, of the steps of any method or process so disclosed.

(45) Furthermore, certain features that are described in this disclosure in the context of separate implementations can also be implemented in combination in a single implementation. Conversely, various features that are described in the context of a single implementation can also be implemented in multiple implementations separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations, one or more features from a claimed combination can, in some cases, be excised from the combination, and the combination may be claimed as a subcombination or variation of a subcombination.

(46) Moreover, while operations may be depicted in the drawings or described in the specification in a particular order, such operations need not be performed in the particular order shown or in sequential order, or that all operations be performed, to achieve desirable results. Other operations that are not depicted or described can be incorporated in the example methods and processes. For example, one or more additional operations can be performed before, after, simultaneously, or between any of the described operations. Further, the operations may be rearranged or reordered in other implementations. Those skilled in the art will appreciate that in some embodiments, the actual steps taken in the processes illustrated and/or disclosed may differ from those shown in the figures. Depending on the embodiment, certain of the steps described above may be removed, others may be added. Furthermore, the features and attributes of the specific embodiments disclosed above may be combined in different ways to form additional embodiments, all of which fall within the scope of the present disclosure. Also, the separation of various system components in the implementations described above should not be understood as requiring such separation in all implementations, and it should be understood that the described components and systems can generally be integrated together in a single product or packaged into multiple products.

(47) For purposes of this disclosure, certain aspects, advantages, and novel features are described herein. Not necessarily all such advantages may be achieved in accordance with any particular embodiment. Thus, for example, those skilled in the art will recognize that the disclosure may be embodied or carried out in a manner that achieves one advantage or a group of advantages as taught herein without necessarily achieving other advantages as may be taught or suggested herein.

(48) Conditional language, such as “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements, and/or steps. Thus, such conditional language is not generally intended to imply that features, elements, and/or steps are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without user input or prompting,

whether these features, elements, and/or steps are included or are to be performed in any particular embodiment.

(49) Conjunctive language such as the phrase “at least one of X, Y, and Z,” unless specifically stated otherwise, is otherwise understood with the context as used in general to convey that an item, term, etc. may be either X, Y, or Z. Thus, such conjunctive language is not generally intended to imply that certain embodiments require the presence of at least one of X, at least one of Y, and at least one of Z.

(50) Language of degree used herein, such as the terms “approximately,” “about,” “generally,” and “substantially” as used herein represent a value, amount, or characteristic close to the stated value, amount, or characteristic that still performs a desired function or achieves a desired result. For example, the terms “approximately,” “about,” “generally,” and “substantially” may refer to an amount that is within less than 10% of, within less than 5% of, within less than 1% of, within less than 0.1% of, and within less than 0.01% of the stated amount. As another example, in certain embodiments, the terms “generally parallel” and “substantially parallel” refer to a value, amount, or characteristic that departs from exactly parallel by less than or equal to 15 degrees, 10 degrees, 5 degrees, 3 degrees, 1 degree, or 0.1 degree.

(51) The scope of the present disclosure is not intended to be limited by the specific disclosures of preferred embodiments in this section or elsewhere in this specification, and may be defined by claims as presented in this section or elsewhere in this specification or as presented in the future. The language of the claims is to be interpreted broadly based on the language employed in the claims and not limited to the examples described in the present specification or during the prosecution of the application, which examples are to be construed as non-exclusive.

(52) Of course, the foregoing description is that of certain features, aspects and advantages of the present invention, to which various changes and modifications can be made without departing from the spirit and scope of the present invention. Moreover, the devices described herein need not feature all of the objects, advantages, features and aspects discussed above. Thus, for example, those of skill in the art will recognize that the invention can be embodied or carried out in a manner that achieves or optimizes one advantage or a group of advantages as taught herein without necessarily achieving other objects or advantages as may be taught or suggested herein. In addition, while a number of variations of the invention have been shown and described in detail, other modifications and methods of use, which are within the scope of this invention, will be readily apparent to those of skill in the art based upon this disclosure. It is contemplated that various combinations or subcombinations of these specific features and aspects of embodiments may be made and still fall within the scope of the invention. Accordingly, it should be understood that various features and aspects of the disclosed embodiments can be combined with or substituted for one another in order to form varying modes of the discussed devices.

Claims

1. A dispenser assembly, comprising: a dispenser bottle and a nipple via which a solution is dispensed, the nipple projecting from a surface of the dispenser bottle; a cap removably couplable over the nipple to a shoulder of the dispenser bottle which circumscribes the surface, the cap comprising: an outer cap; an inner cap extending within the outer cap; and a seal at or proximal an end of the inner cap; and an eye cup fixedly coupled to the surface of the dispenser bottle and configured to fit within the cap when the cap is coupled to the dispenser bottle, the eye cup comprising: a circumferential wall disposed around the nipple; and a distal edge of the circumferential wall defining an oval shape, the distal edge configured to be disposed over an eye socket to facilitate delivery of a solution into a user's eye, wherein the nipple is disposed between the distal edge of the circumferential wall and the surface, and wherein the circumferential wall tapers continuously outward from the surface to the distal edge, wherein when the cap is coupled to

- the dispenser bottle, the seal sits adjacent the nipple to inhibit leakage of solution from the nipple and wherein the eye cup fits within the outer cap when the cap is coupled to the dispenser bottle.
2. The dispenser assembly of claim 1, wherein the nipple extends within the outer cap when the cap is coupled to the dispenser bottle.
 3. The dispenser assembly of claim 1, wherein the seal is one piece with the inner cap and comprises rubber or plastic.
 4. The dispenser assembly of claim 1, wherein the dispenser assembly is configured to prevent contamination of the nipple and prevent inadvertent contact of the nipple with a user's eye.
 5. The dispenser assembly of claim 1, wherein the distal edge of the eye cup is contoured to fit around a periorbital region.
 6. The dispenser assembly of claim 1, wherein a central axis of the eye cup is aligned with a central axis of the dispenser bottle.
 7. The dispenser assembly of claim 1, wherein a height of the nipple above the dispenser bottle is less than half a height of the distal edge of the eye cup above the dispenser bottle.
 8. The dispenser assembly of claim 1, wherein the cap is couplable to the shoulder with an annular snap fit.
 9. A dispenser assembly, comprising: a dispenser bottle and a nipple via which a solution is dispensed, the nipple projecting from a surface of the dispenser bottle; a cap removably couplable over the nipple to the surface, the cap comprising: an outer cap; an inner cap extending within the outer cap; and a seal at or proximal an end of the inner cap; and an eye cup fixedly coupled to the surface of the dispenser bottle and configured to fit within the cap when the cap is coupled to the dispenser bottle, the eye cup comprising: a circumferential wall disposed around the nipple; and a distal edge of the circumferential wall defining an oval shape, the distal edge configured to be disposed over an eye socket to facilitate delivery of a solution into a user's eye, wherein the nipple is disposed between the distal edge of the circumferential wall and the surface, and wherein the circumferential wall tapers continuously outward from the surface to the distal edge, wherein when the cap is coupled to the dispenser bottle, the seal sits adjacent the nipple to inhibit leakage of solution from the nipple and wherein the eye cup fits within the outer cap when the cap is coupled to the dispenser bottle.
 10. The dispenser assembly of claim 9, wherein the eye cup is configured to inhibit the nipple from touching the user's eye.
 11. The dispenser assembly of claim 9, wherein the dispenser bottle is formed from a resilient material.
 12. The dispenser assembly of claim 9, wherein a distal portion of the inner cap extends past the seal, the distal portion of the inner cap configured to encircle the nipple when the cap is coupled to the dispenser bottle.
 13. The dispenser assembly of claim 9, wherein a distal portion of the inner cap is couplable to the nipple.
 14. A dispenser assembly, comprising: a dispenser bottle and a nipple via which a solution is dispensed, the nipple projecting from a surface of the dispenser bottle; a cap removably couplable over the nipple to the surface, the cap comprising: an outer cap; an inner cap extending within the outer cap; and a seal at or proximal an end of the inner cap; and an eye cup fixedly coupled to the surface of the dispenser bottle and configured to fit within the cap when the cap is coupled to the dispenser bottle, the eye cup comprising: a circumferential wall disposed around the nipple; and a distal edge of the circumferential wall defining an oval shape, the distal edge configured to be disposed over an eye socket to facilitate delivery of a solution into a user's eye, wherein the nipple is disposed between the distal edge of the circumferential wall and the surface, and wherein the circumferential wall tapers continuously outward from the surface to the distal edge, wherein when the cap is coupled to the dispenser bottle, the seal sits adjacent the nipple to inhibit leakage of solution from the nipple.

15. The dispenser assembly of claim 14, wherein the eye cup is configured to inhibit the nipple from touching the user's eye.
 16. The dispenser assembly of claim 14, wherein a height of the nipple above the surface is less than half a height of the distal edge of the eye cup above the surface.
 17. The dispenser assembly of claim 14, wherein the dispenser bottle has a flat base surface.
 18. The dispenser assembly of claim 14, wherein a distal portion of the inner cap is couplable to the nipple.
 19. The dispenser assembly of claim 14, wherein a central axis of the eye cup is aligned with a central axis of the dispenser bottle.
 20. The dispenser assembly of claim 14, wherein the nipple extends within the outer cap when the cap is coupled to the dispenser bottle.
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